

W6D4 RAG Pipeline

Retrieval-Augmented Generation, or RAG, is a technique used in artificial intelligence applications.

RAG combines information retrieval with language generation. First, documents are divided into smaller chunks. These chunks are converted into numerical embeddings and stored in a vector database.

ChromaDB is a vector database that can store embeddings and retrieve information based on semantic similarity.

Cosine similarity is commonly used to compare embeddings. It measures how similar two vectors are based on the angle between them.

Ollama allows large language models to run locally. In this project, Ollama is used for both text embeddings and language generation.

The nomic-embed-text model is used to create embeddings. The llama3.2:3b model is used to generate answers.

The RAG pipeline works as follows:

1. Load the PDF.
2. Extract text from the PDF.
3. Split the text into chunks.
4. Create embeddings for the chunks.
5. Store embeddings in ChromaDB.
6. Convert the user's question into an embedding.
7. Retrieve the three most relevant chunks.
8. Send the retrieved context to the language model.
9. Generate the final answer.

RAG is useful because it allows language models to answer questions using information from a specific knowledge source.